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Asymmetric Catalysis

Catalytic Enantioselective Friedel–Crafts Allenylation

*Priyotosh Das, Debangshu Ghosh, and Santanu Mukherjee**

Abstract: The first enantioselective Friedel-Crafts (FC) allenylation reaction for the creation of central chirality is developed under cooperative Ir(I)/(phosphoramidite,olefin) and Lewis acid catalysis. This enantioconvergent reaction utilizes racemic allenylic alcohol as the electrophile and shows compatibility with a variety of electron-rich arenes and heteroarenes. The resulting highly enantioenriched (up to >99.5:0.5 e.r.) 1,1-disubstituted allenylic methanes, bearing a benzylic carbon stereocenter, are obtained with complete regiocontrol – both on (hetero)arenes as well as on the allenylic fragment. This protocol allows for the enantioselective formal introduction of a 4-carbon alkyl chain into (hetero)arenes, along with the creation of a benzylic stereocenter. Judicious synthetic elaborations not only lead to formal enantioselective FC alkylation products of less electron-rich arenes but also of substituted arenes in *ortho*- and even *meta*-selective fashion. An intramolecular version of this FC allenylation is shown to proceed with promising enantioselectivity under the same catalytic conditions. Mechanistic studies revealed the involvement of dynamic kinetic asymmetric transformation (DyKAT) of racemic allenylic alcohols in this reaction.

Introduction

Enantioselective introduction of an unfunctionalized alkyl group into (hetero)arenes holds significant importance owing to the prevalence of benzylic carbon stereocenters in natural products and medicinally active compounds.[1] Notwithstanding the magnificent insurgence of transition metal-catalyzed C–H activation chemistry during the past decade,[2] Friedel–Crafts (FC) alkylation[3] remained arguably the most straightforward strategy for the introduction of unfunctionalized alkyl groups into (hetero)arenes (Scheme 1A).[4]

Over the years, a plethora of catalytic methods have been instituted to achieve enantioselective FC reactions.[5] An overwhelming majority of these reactions relied on using substrates bearing electrophilic sp^2 carbon center.[5] In addition to being prochiral, functional groups present in these substrates ease their activation and enantiofacial discrimination (Scheme 1B). In contrast, the application of electrophiles containing a reactive sp^3 carbon center in enantioselective FC reactions has been scarce.[6] Such racemic electrophilic substrates necessitate the intervention of either a dynamic kinetic resolution (DKR) or a dynamic kinetic asymmetric transformation (DyKAT) pathway to avoid using an overstoichiometric amount of electrophile.[7] For reactions proceeding via a prochiral intermediate (e.g., carbocation), discrimination between the two enantiotopic faces becomes an enormous challenge due to the absence of any functional group (Scheme 1B).[8]

Transition metal-catalyzed allylic[9–11] and allenylic substitutions[11a] are transformations that use electrophiles bearing sp^3 carbon at the reaction center. Several inter- and intramolecular enantioselective FC allylation reactions have been developed under Pd[12] and Ir-catalysis[13]. However, the closely related FC allenylation reaction, which would formally introduce an unfunctionalized 4-carbon alkyl chain into (hetero)arenes, with the generation of a benzylic carbon stereocenter, remained largely unexplored (Scheme 1B)

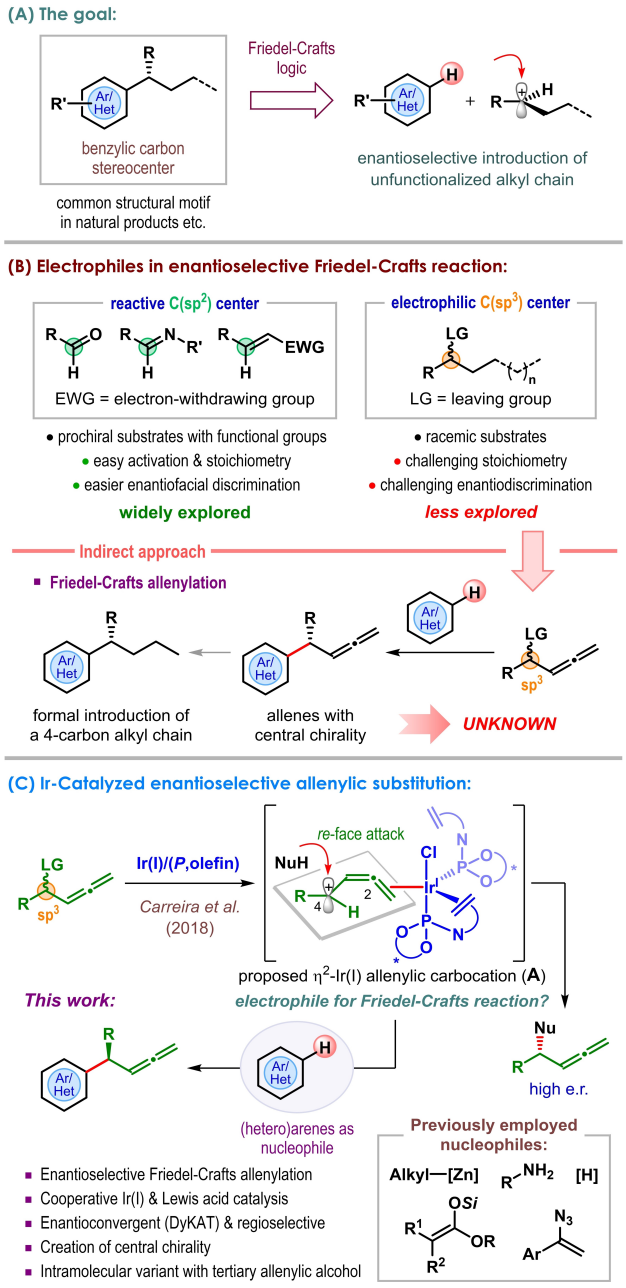
During this study, a report by Yu, Shao, and co-workers described a Pd-catalyzed FC allenylation of indoles for the enantioselective synthesis of axially chiral allenes.[14] However, enantioselective FC allenylation for creating central chirality remains an uncharted territory.

Iridium-catalyzed enantioselective allenylic substitution reaction, introduced by Carreira *et al.* in 2018,[15] is mechanistically distinct from its more widely explored palladium-catalyzed variants.[16,17] Unlike the involvement of η^3-Pd(II)-butadienyl intermediates in Pd(0)-catalyzed allenylic substitution,[16,17] the reactions catalyzed by Ir(I)/(phosphoramidite,olefin)-complex[11c] is proposed to proceed through a carbocationic η^2-Ir(I) intermediate **A** (Scheme 1C),[15] which makes it a potential candidate for FC reaction. However, (hetero)arenes are a class of nucleophiles which has never been employed in this reaction. As such, only a few nucleophiles have so far been explored in this Ir-catalyzed enantioselective allenylic substitution reaction, namely alkylzinc reagents,[15] ammonia surrogates,[18] hydride (from Hantzsch ester),[19] silyl ketene acetals[20] and vinyl azides[21] (Scheme 1C).

As a part of our ongoing research program dedicated to the development of enantioselective C(sp^2)–H allylation[22] and allenylation reactions, we decided to employ (hetero)arenes as the nucleophile in Ir-catalyzed allenylation reaction. However, (hetero)arenes are less reactive compared to the previously utilized nucleophiles in this reaction (see above).[15,18–21] Nonetheless, we reasoned that the proposed η^2-Ir(I)-bound allenylic carbocation would be sufficiently electrophilic to be intercepted by suitably

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Scheme 1. Enantioselective Friedel–Crafts allenylation: the backdrop of this study.

substituted electron-rich (hetero)arenes and should facilitate the desired FC reaction. Moreover, the proposed trigonal bipyramidal coordination sphere of Ir(I) bearing two (*P*,olefin) ligands, even though distant from the reaction site, has been shown to offer necessary face-selectivity for nucleophilic addition to the prochiral carbocation in **A** (Scheme 1C).[15,19] Therefore, the same selectivity can be expected in the enantiodetermining C–C bond formation step with (hetero)arenes.

Apart from enantioselectivity, we were cognizant of two regiochemical issues: While the regioselectivity on the suggested allenylic carbocation **A** has been shown to be reliably dictated in favor of nucleophilic addition at C4 (vs C2) by the ligands on Ir (Scheme 1C),[15] the site selectivity on the (hetero)arenes must be left to their innate reactivity.

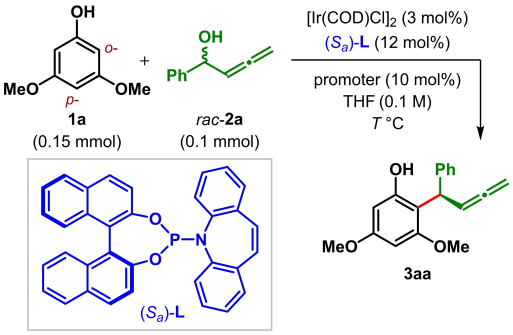
Herein, we present the results of this investigation, which culminated in the first Ir-catalyzed enantioselective Friedel–Crafts allenylation (Scheme 1C).

Results and Discussion

At the outset of our study, FC allenylation of 3,5-dimethoxyphenol **1a** with racemic allenylic alcohol *rac*-**2a** in the presence of 6 mol % of a catalyst derived from $[Ir(COD)Cl]_2$ and Carreira's (*P*,olefin) ligand (S_a)-**L**[11c] was selected as the model reaction (Table 1).

The reason behind the choice of this substrate was twofold. While the electron-rich nature of **1a** was reasoned to facilitate the FC reaction, we were apprehensive of the possible overalkylation processes. Therefore, it was essential to identify reaction conditions that would favor only the monoalkylated product. At the same time, we wanted to verify the regioselectivity of the monoalkylation process (i.e., *ortho*- vs *para*-substitution) with an unsymmetrical arene such as **1a**.

Table 1: Optimization of reaction conditions for enantioselective FC allenylation.[a]



Entry	Promoter	*T* (°C)	*t* [h]	Yield [%][b]	e.r.[c]
1	$Sc(OTf)_3$	50	3	38	78:22
2	$Bi(OTf)_3$	50	3	<5	n.d.
3	$Fe(OTf)_2$	50	3	(54)	97.5:2.5
4	$Zn(OTf)_2$	50	3	55	98:2
5	$Yb(OTf)_3$	50	3	(55)	96:4
6	$La(OTf)_3$	50	3	58 (52)	98:2
7	$BF_3{\cdot}OEt_2$	50	3	29	81:19
8	CSA	50	3	55	93:7
9[d]	$Fe(OTf)_2$	50	3	53	85:15
10[e]	$Fe(OTf)_2$	50	3	77 (73)	95.5:4.5
11[e]	$Fe(OTf)_2$	25	18	77	95:5
12[f]	$Fe(OTf)_2$	25	18	(71)	98:2
13[f]	$La(OTf)_3$	25	24	59 (54)	82:18

[a] Unless noted otherwise, reactions were carried out on a 0.1 mmol scale. [b] Yields were determined by ^1H NMR analysis with mesitylene as internal standard. Isolated yield is given in parentheses. [c] Enantiomeric ratios (e.r.) were determined by HPLC analysis on a chiral stationary phase. [d] Reaction in CH_2Cl_2. [e] Reaction in toluene. [f] Reaction in 1:1 toluene/THF. CSA = camphorsulfonic acid. n.d. = not determined.

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The reaction with an excess of arene **1a** in THF at 50 °C using $Sc(OTf)_3$ as the Lewis acidic promoter indeed furnished the FC allenylation product **3aa** (Table 1, entry 1). Despite the low yield and modest enantioselectivity, the formation of only the monoallenylation product with complete regioselectivity in favor of *ortho*- was encouraging and provided the impetus for further investigation. In line with earlier studies,[15,18–21] no diene product, arising from the addition of **1a** at C2 of **A** (Scheme 1C), could be detected. Screening of other Lewis and Brønsted acidic promoters (entries 2–8) revealed $Fe(OTf)_2$ and $La(OTf)_3$ to be nearly equally efficient as **3aa** was isolated with similar levels of yield and enantiomeric ratio (e.r.) in both these cases (entries 3 and 6). Notably, a stronger Lewis acid $Bi(OTf)_3$ led to the complete decomposition of **2a**, and no **3aa** could be detected (entry 2). At this stage, we decided to use $Fe(OTf)_2$ for further optimization. The reaction in toluene was found to be faster, but **3aa** was formed with diminished e.r. (entry 10). In this reaction, no allenylation product of toluene could be detected despite its presence in large excess, which underscores the importance of electron-rich substituents on the arene ring in rendering it nucleophilic (see below). It is possible to carry out this reaction at room temperature without much influence on the enantioselectivity (entry 11). Although expectedly slower, the reaction at room temperature makes the process operationally simpler. Combining the effect of toluene in improving the reaction rate and that of THF in enhancing the enantioselectivity led to the identification of a 1:1 mixture of THF and toluene as the optimum reaction medium when **3aa** was isolated in 71 % yield with 98:2 e.r. (entry 12). Under the same reaction conditions, $La(OTf)_3$ delivered **3aa** with a much inferior yield and enantioselectivity (entry 13). Efforts to ameliorate the reaction outcome by tweaking other reaction parameters proved unsuccessful.[23]

The catalyst, promoter, and the reaction conditions optimized for the FC allenylation of 3,5-dimethoxyphenol **1a** (Table 1, entry 12) were then extended to other arenes and heteroarenes for reaction with racemic allenylic alcohol *rac*-**2a** (Scheme 2). We were delighted that our protocol is not restricted to electron-rich phenols such as **1a**. Structurally and electronically related arenes such as 1,3,5-trimethoxybenzene **1b** and 3-methoxy-*N*,*N*-dimethylaniline **1c** fared equally well under our reaction conditions and afforded allenylic arenes **3ba** and **3ca**, respectively, with similar level of yields and enantioselectivities (Scheme 2A). With **1c** as the substrate, allenylation took place exclusively at *para*- to dimethylamino group. The presence of hydroxy or methoxy group on the arene ring is not a prerequisite for this reaction. Replacing the methoxy group in **1c** with a less electron-releasing methyl doesn't appear to affect the reactivity of **1d**, as the regioselective *para*-product **3da** was isolated in 84 % yield and with 99:1 e.r. In fact, FC allenylation of *N*,*N*-dimethylaniline (**1e**) and other analogous *N*,*N*-dialkylaniline derivatives (**1f–i)** themselves were found to occur efficiently and selectively at the *para*-position of the *N*,*N*-dialkylamine substituent in moderate to good yield with good to excellent enantioselectivity. Naphthalene derivatives (**1j–l)**, adorned with electron-rich substituents, are viable substrates for this allenylic arylation reaction. However, these reactions turned out to be much less efficient and enantioselective. Unfortunately, electronically lesser endowed arenes such as phenol, anisole, and naphthol (both α- and β-) failed to react under our optimized conditions.[23]

In the realm of heteroarenes, indole (**1m**) as well as *N*-substituted indoles (**1n–o)** served as excellent substrates and furnished 3-allenylic indoles **3ma–oa** in very high yield with superb enantioselectivity (Scheme 2B). The reaction with 2-methylindole (**1p**), although somewhat sluggish, was equally enantioselective. In the case of 3-methylindole (**1q**), 2-allenylic indole **3qa** was formed in 73 % yield with 98:2 e.r. Whether this reaction proceeds by direct 2-allenylation or via initial 3-allenylation of indole followed by stereospecific rearrangement could not be confirmed at this stage. Pyrrole (**1r**) and its derivatives (**1s–u)** showed fairly good reactivities, even though enantioselectivities varied depending on the nature and the position of the substituents. While pyrrole (**1r**) itself and *N*-phenylpyrrole (**1s**) afforded the 2-allenylation products **3ra** and **3sa**, respectively, with excellent enantioselectivities, the reactions of 2-methylpyrrole (**1t**) and particularly 2,4-dimethylpyrrole (**1u**) took place with significantly reduced enantioselectivity. 3-Methoxythiophene (**1v**), on the other hand, turned out to be a suitable substrate and delivered 2-allenylic thiophene derivative **3va** selectively with an outstanding e.r. of $>99.5:0.5$. Among other heteroarenes tested, (benzo)furans, carbazoles, and *N*-methylimidazole were found to be unsuitable substrates for FC allenylation under our optimized protocol, presumably due to their reduced nucleophilicity.[23]

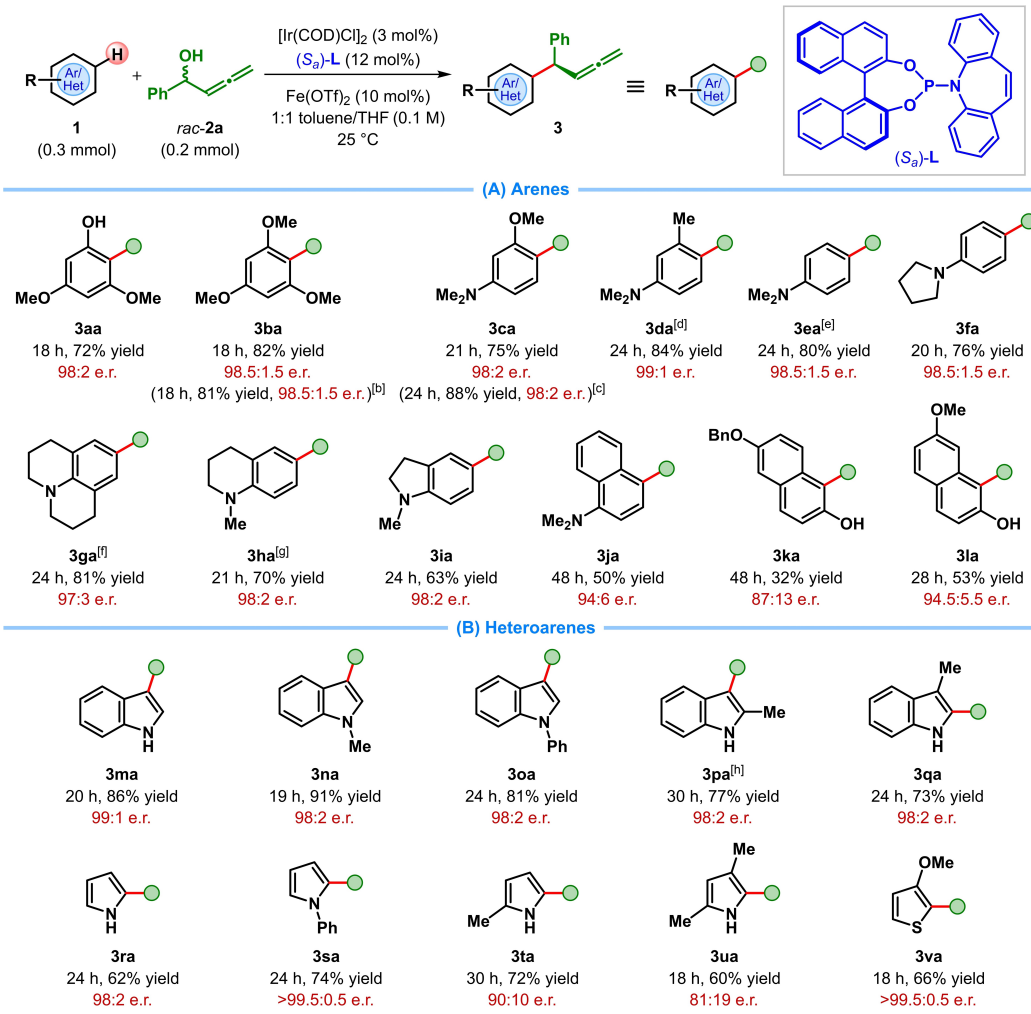
For showcasing the scope of allenylic alcohols, 1,3,5-trimethoxybenzene (**1b**) was chosen as the nucleophile. The examples compiled in Scheme 3A indicate the generality of this enantioselective FC allenylation protocol with respect to racemic allenylic alcohols bearing aryl substituents. Electronically diverse substituents on various positions of the aryl ring were well tolerated and furnished the desired products generally in high yield and with excellent enantioselectivity. Of specific note is the FC product **3bk**, which bears a boronate functionality and could be subjected to diversification through various cross-coupling reactions. The enantiopurity of 1-bromonaphthyl-substituted product **3bp** could be further improved after a single crystallization. Single crystal X-ray diffraction analysis of crystals thus obtained for **3bp** established its absolute configuration to be (*S*) (CCDC 2302758, Scheme 3A).[24] Configuration of the other FC allenylation products shown in this report were assigned in analogy with **3bp**.

Apart from aryl substituents, as shown in Scheme 3A, allenylic alcohols containing heterocyclic substituents, such as 1,3-benzodioxole, indole, furan, and thiophene, could also be utilized as allenylic electrophile for reaction with **1b** (Scheme 3B)

However, most of these reactions were relatively less enantioselective than their aryl counterparts. While most aliphatic allenylic alcohols failed to react under our optimized conditions,[23] cyclopropyl-substituted allenylic alcohol **2v** displayed substantial reactivity and furnished the

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[a] Unless noted otherwise, reaction conditions indicated above were followed. Yields correspond to the isolated yield after chromatographic purification. Enantiomeric ratios (e.r.) were determined by HPLC analysis using a stationary phase chiral column. [b] Reaction on a 1.0 mmol scale. [c] Reaction using 1.0 mmol of **1c** and 1.2 mmol of *rac*-**2a** under 4 mol% of Ir[(S_a)-**L**]$_2$Cl. [d] Reaction using 0.5 mmol of **1d** and 0.75 mmol of *rac*-**2a** under 4 mol% of Ir[(S_a)-**L**]$_2$Cl. [e] Reaction using 0.4 mmol of **1d** and 0.6 mmol of *rac*-**2a**. [f] Reaction using 0.4 mmol of **1g** and 0.6 mmol of *rac*-**2a**. [g] Reaction using 0.5 mmol of **1h** and 0.75 mmol of *rac*-**2a**. [h] Reaction using 0.24 mmol of *rac*-**2a** and 0.2 mmol of **1p**.

Scheme 2. Scope of (hetero)arenes in enantioselective FC allenylation.[a]

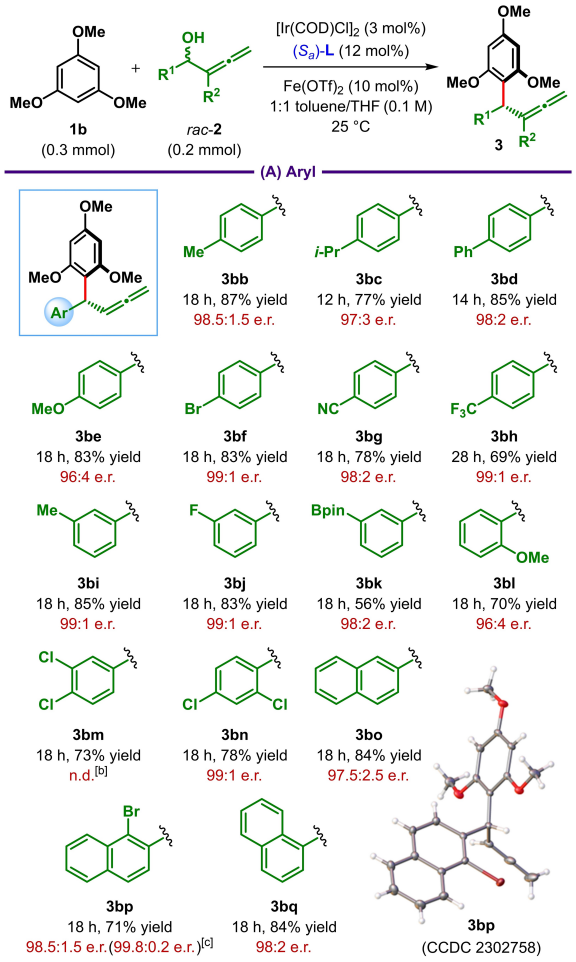
desired product **3bv** in 88% yield, albeit with only modest enantioselectivity. However, no FC allenylation product could be detected when tertiary allenylic alcohol was used as the electrophilic partner for reaction with **1b**, even at 50 °C.[23]

In the Ir-catalyzed enantioselective allenylic substitution reactions reported to date, allenylic alcohols or their derivatives, bearing monosubstituted allene unit, have most commonly been employed as the allenylic electrophile. Reactions with allenylic alcohols, having disubstituted allene unit, have been shown to be less efficient and considerably less enantioselective.[15,18] We were pleased to find 2,2-disubstituted allenylic alcohol **2w** to be a competent substrate for our FC allenylation reaction. However, as expected, **2w** was found to be less reactive compared to its monosubstituted analogue **2a**. Accordingly, the reaction needed to be carried out at an elevated temperature (50 °C), and the product **3bw** was obtained with moderate e.r.

Having showcased the broad generality of both the reaction partners in this enantioselective FC allenylation reaction, we recognized the potential of our protocol in providing access to both the product enantiomers under the influence of a single catalyst antipode – simply through the choice of substrates. The implementation of this strategy was demonstrated by enantiodivergent synthesis of diarylallenylic methane **3es** (Scheme 4). The reaction of *N*,*N*-dimethylaniline **1e** with *N*-tosyl-3-indolyl substituted allenylic alcohol **2s** under our standard conditions using (S_a)-(*P*,olefin) ligand (S_a)-**L** afforded (*S*)-**3es** in 62% yield with 97:3 e.r. On the other hand, when indole (**1m**) was

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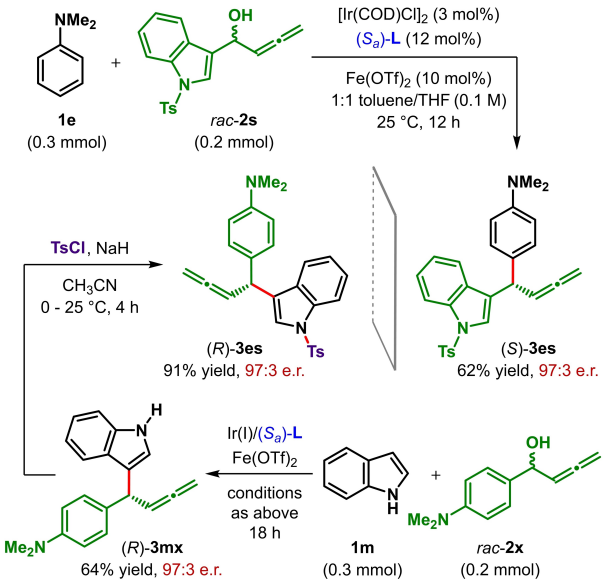
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[a] Unless noted otherwise, reaction conditions indicated above were followed. Yields correspond to the isolated yield after chromatographic purification. Enantiomeric ratios (e.r.) were determined by HPLC analysis using a stationary phase chiral column. [b] n.d. = e.r. could not be determined with the available chiral columns. [c] The numbers in the parentheses indicate e.r. after a single recrystallization. [d] Reaction using 0.2 mmol of **1b** and 0.3 mmol of *rac*-**2t**. [e] Reaction using 0.2 mmol of **1b** and 0.3 mmol of *rac*-**2v**. [f] Reaction at 50 °C.

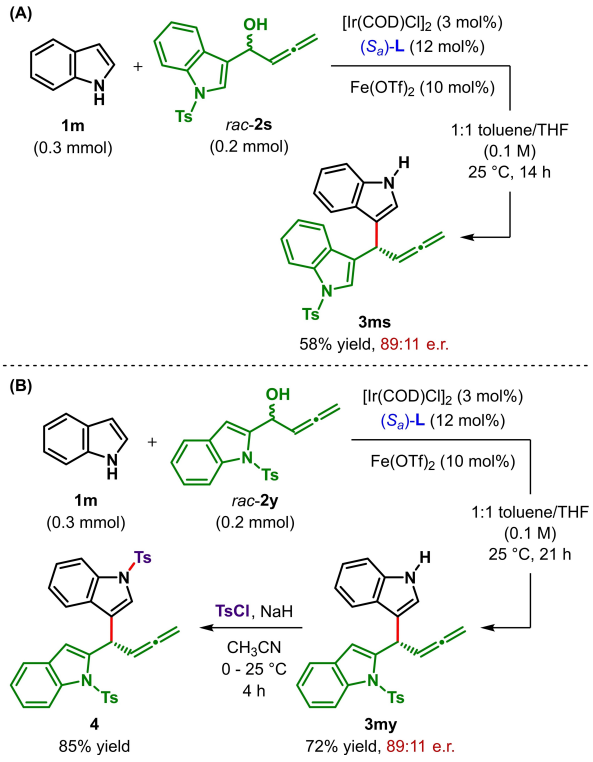
Scheme 3. Scope of allenylic alcohols in enantioselective FC allenylation.[a]

employed as the nucleophile for reaction with *N*,*N*-dimethylaniline-substituted allenylic alcohol **2x** under the same catalytic conditions, (*R*)-**3mx** was obtained in 64% yield with 97:3 e.r. *N*-Tosylation of indole in (*R*)-**3mx** resulted in (*R*)-**3es** with complete conservation of stereochemical integrity.



Scheme 4. Enantiodivergent synthesis of diarylallenylic methane **3es**.

Our protocol also provides a platform for the enantioselective synthesis of chiral allenylic methane derivatives bearing two similar substituents at the stereocenter. For example, FC allenylation of indole (**1m**) with *N*-tosyl-3-indolyl substituted allenylic alcohol (**2s**) delivered bis-(indolyl)allenylic methane **3ms** with 89:11 e.r. (Scheme 5A).



Scheme 5. Enantioselective syntheses of bis(indolyl)allenylic methanes.

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Stereogenicity of **3ms** arises due to the *N*-protection of one of the indole moieties. Similarly, the reaction of indole (**1m**) with *N*-tosyl-2-indolyl substituted allenylic alcohol (**2y**) furnished another bis(indolyl)allenylic methane **3my** with 89:11 e.r. (Scheme 5B). *N*-Tosylation of **3my** generated **4**, which is chiral due to the differential connectivity of the two indole rings (C2 and C3) with the stereocenter. Enantioselective synthesis of both **3ms** and **4** could be difficult by other means.

As shown in Scheme 2, indoles (**1m–o**) and indoline (**1i**) display different regioselectivity in this FC reaction. This innate reactivity pattern of indole and its derivatives was exploited for the regiodivergent allenylation of indoles (Scheme 6). While direct FC reaction of *N*-methylindole (**1n**) led to 3-allenylic indole **3na**, a two-step protocol consisting of FC allenylation of *N*-methylindoline (**1i**)[25] followed by oxidation with MnO_2[13b] resulted in 5-allenylic indole **5** in 47% overall yield with 98:2 e.r. Similarly, starting from 4,7-dihydro-1*H*-indole (**1w**), FC allenylation and oxidative aromatization with *p*-benzoquinone[26] furnished 2-allenylic indole **6**. Unfortunately, **6** was found to be sensitive to silica gel and configurationally labile.

Apart from the 1.0 mmol scale reactions depicted for **3ba** and **3ca** in Scheme 2, this enantioselective FC allenylation reaction can be further scaled up with lower catalyst loading. As exemplified for the reaction between **1a** and *rac*-**2a** in the presence of 4 mol% *in situ* generated Ir[(S_a)-**L**]$_2$Cl, both the catalytic activity as well as the yield and enantioselectivity were retained even with a 20-fold increase in the scale of the reaction (Scheme 7A).

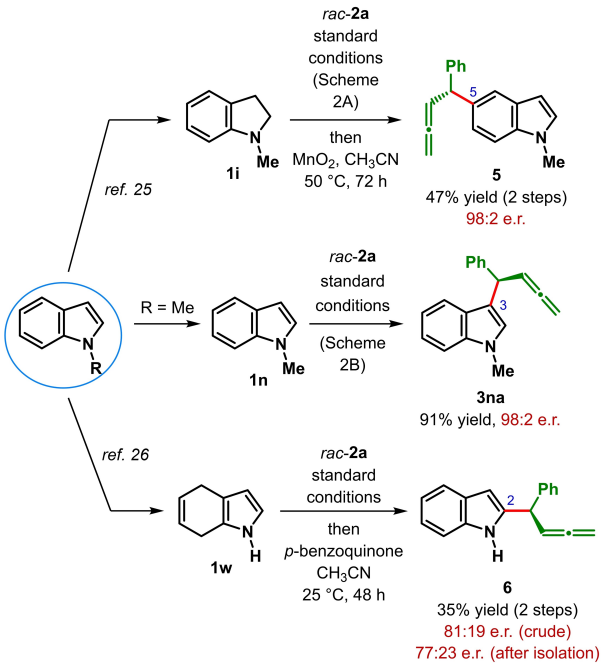
The allene moiety in allenylic arenes can act as a linchpin for synthetic elaboration to other building blocks, especially in cooperation with functionalities at the *ortho*-position of the arene ring. Molybdenum-catalyzed hydrosilylation of allene in **3ba** proved to be highly regio- and diastereoselective and resulted in (*Z*)-allylsilane **7** in 61% yield (Scheme 7B).[27] Rhodium-catalyzed intramolecular hydroalkoxylation of **3aa** gave dihydrobenzofuran derivative **8** exclusively as the *cis*-isomer in 84% yield.[28] The corresponding *trans*-dihydrobenzofuran **9** can be obtained under gold-catalysis,[29] but with considerably less d.r. Organocatalytic intramolecular iodoetherification[30] of **3aa** worked reasonably well under bifunctional tertiary aminothiourea ***epi*-QD-TU** to give 2-iodovinyl dihydrobenzofuran derivative **10**, albeit with only 2.2:1 d.r.

Hydrogenation of the allene functionality allows for the formal introduction of an unfunctionalized 4-carbon alkyl chain into a (hetero)arene ring with the creation of a benzylic carbon stereocenter as shown for the conversion of **3aa** to **11** (Scheme 7B). The presence of the hydroxy group in **11** provides a handle for installing other functional groups on the arene ring through the intermediacy of the corresponding triflate **12**. For instance, Pd-catalyzed Suzuki coupling of **12** with phenylboronic acid gave biaryl derivative **13** in excellent yield. Another Pd-catalyzed coupling with diphenylphosphine oxide led to **14** through the formation of a C–P bond. The triflate group in **12** can also be removed under Pd-catalyzed hydrogenolytic conditions using triethylammonium formate[31] to furnish the formal FC alkylation product of 1,3-dimethoxybenzene **15** – directly inaccessible via FC allenylation reaction due to insufficient nucleophilicity of 1,3-dimethoxybenzene.

The use of *N*,*N*-dialkylaniline derivatives as a nucleophile in the FC allenylation reaction is equally advantageous for accessing formal FC alkylation products of less electron-rich arenes. Cleavage of the dialkylamino substituents from arenes is possible following the two-step protocol developed by MacMillan et al.[32] Thus, allenylic arenes **3ca** and **3da** could be transformed into formal FC alkylation products of anisole (**17**) and toluene (**19**), respectively, in excellent yield (Scheme 7C, eq 1). This synthetic maneuver not only allowed for the access of FC alkylation products of inherently inert nucleophiles, such as toluene and anisole, in the Ir-catalyzed FC allenylation reaction (see above) but also ensures the otherwise challenging '*ortho*-selectivity' in the products **17** and **19**.

The exposure of 6-allenylic *N*-methyl tetrahydroquinoline **3ha** to this three-step protocol resulted in the rupture of the tetrahydroquinoline ring to form **21** (Scheme 7C, eq 2). This strategy presents a facile and enantioselective route to '*meta*-selective' formal FC alkylation products through the utilization of cyclic *N*,*N*-dialkylaniline derivatives as the nucleophile. Starting from the tricyclic substrate **3ga**, the same synthetic manipulation led to *meta*,*meta*-disubstituted formal FC alkylation product **23**, with the concomitant formation of a 10-membered macrocycle (Scheme 7C, eq 3).

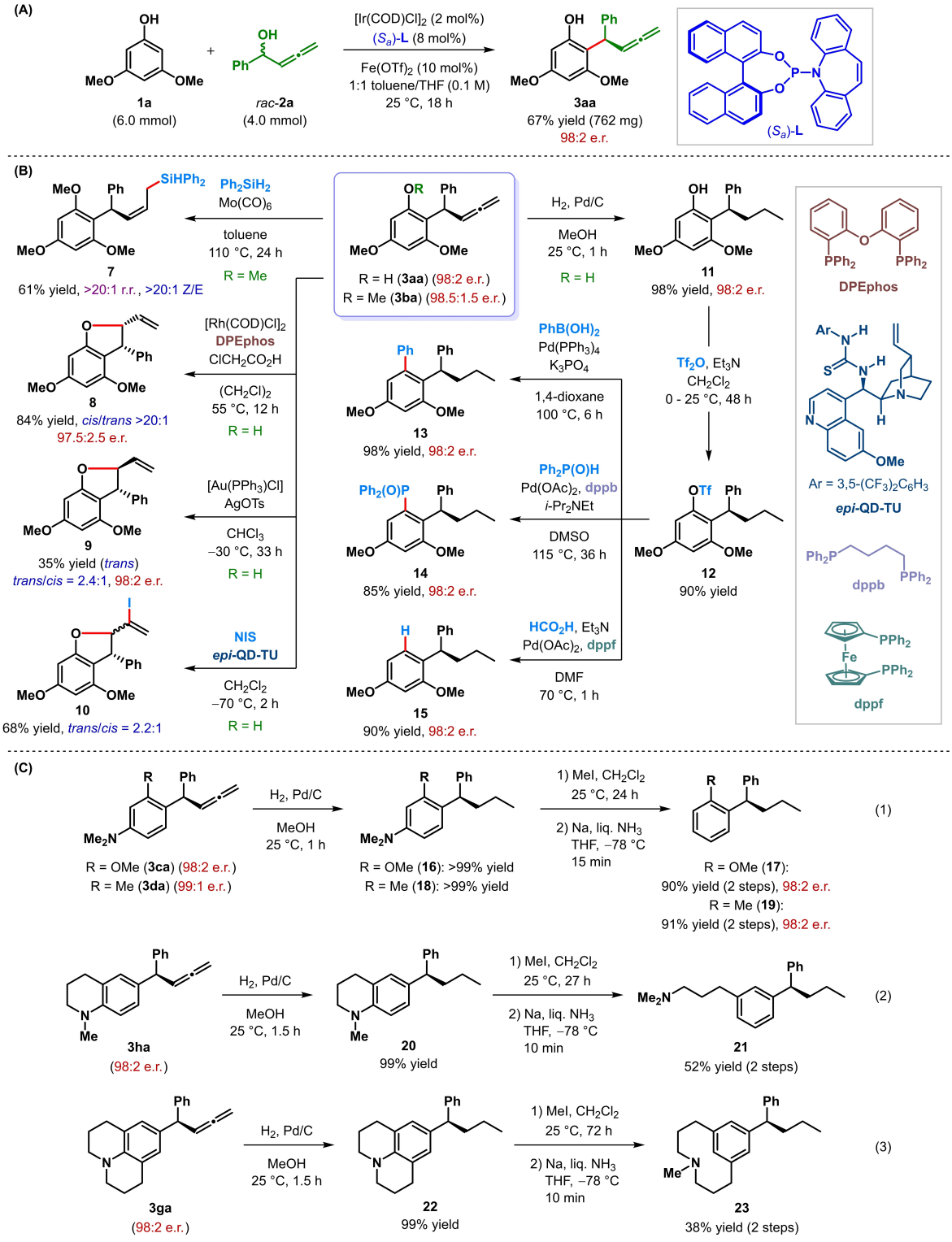
Despite notable progress in asymmetric allenylic substitution reactions under Pd and Ir-catalysis, there has been no report on their intramolecular variants.[33] After exploring the scope and limitations of this Ir-catalyzed enantioselective FC allenylation reaction, we wondered about the viability of the same catalyst system for an intramolecular



Scheme 6. Regiodivergent allenylation of indoles.

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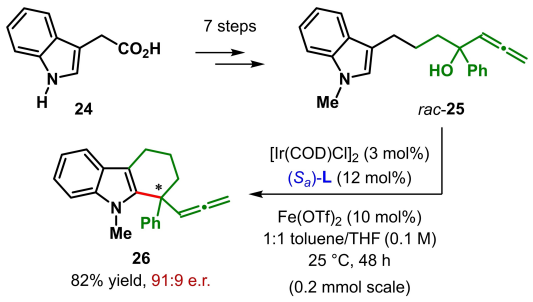


Scheme 7. Scale-up of enantioselective FC allenylation reaction and synthetic elaboration of products.

variant. For this purpose, indole-tethered racemic tertiary allenylic alcohol *rac*-**25** was synthesized from commercially available indole-3-acetic acid **24** (Scheme 8). When subjected to our standard reaction conditions, *rac*-**25** underwent facile intramolecular FC reaction to furnish tetrahydrocarbazole derivative **26**, bearing an all-carbon quaternary stereogenic center, in 82 % yield with 91:9 e.r. This is the first example of an enantioselective intramolecular allenylic substitution reaction with a carbon nucleophile.

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Scheme 8. Ir-Catalyzed enantioselective intramolecular Friedel-Crafts allenylation.

Based on the mechanistic studies carried out by the Carreira group[15,19] and our earlier experimental findings,[21] a plausible catalytic cycle for this FC allenylation reaction, as shown in Scheme 9A, can be envisaged. The reaction is believed to be initiated by the coordination of the allenylic alcohol **2a** to the *in situ* generated Ir(I)[(S_a)-**L**]$_2$ complex via its terminal double bond in an η^2-fashion. Since this coordination of catalyst with racemic allenylic alcohol gives rise to two diastereomeric complexes (R,S_a,S_a)-**27** and (S,S_a,S_a)-**28**, preferential formation of one of these two complexes presents a possibility, which can have further consequences. To get insight into this complexation step, we monitored the formation of the (η^2-allenylic alcohol)Ir(I) complexes using $^{31}P\{^1H\}$ NMR spectroscopy (Scheme 9B). Combining [Ir(cod)Cl]$_2$, (S_a)-**L** and *rac*-**2a** in 1:4:10 ratio in toluene-d$_8$ produced a few species at equilibrium, among which the two distinct sets of doublets were noteworthy: 104.2 and 116.6 ppm ($^2J_{P\text{-}P}$=22.3 Hz); 104.6 and 115.6 ppm ($^2J_{P\text{-}P}$=22.4 Hz).[19] Independent $^{31}P\{^1H\}$ NMR studies using enantioenriched **2a** confirmed these doublets correspond to (S,S_a,S_a)-**28** and (R,S_a,S_a)-**27**, respectively. These NMR data also supported the presence of two nonequivalent phosphorus donors at the iridium center. Notably, the ratio of these two complexes was found to be ~1:1, which ruled out the possibility of preferential coordination of one of the enantiomers of **2a** to the Ir-center, likely because of the long distance between the stereocenters on **2a** and iridium.

Ionization of these η^2-complexes in the presence of a Lewis acidic promoter has been proposed to generate two diastereomeric Ir-bound butadienylium carbocations.[15] To understand the nature of the reactive intermediate in this FC reaction, an intermolecular competition experiment was conducted using an equimolar mixture of an electron-rich and an electron-poor allenylic alcohol **2e** and **2h**, respectively (Scheme 9C). With 1,3,5-trimethoxybenzene (**1b**) as the nucleophile under our standard catalytic conditions, **2e** was found to react faster than **2h**. The higher reactivity of **2e** can be ascribed to the greater stabilization of the intermediate and the subsequent transition state having carbocationic character.

The ionization step (i.e., the conversion of **27** and **28**, respectively, to **29** and **30**) is expected to be stereospecific and hence sow the seed for a potential kinetic resolution (KR) of allenylic alcohol. However, the use of allenylic alcohols as the limiting reagent in our enantioselective FC reaction and isolation of products with excellent e.r. in >50 % yield contradict this possibility. Moreover, the reaction between **1a** and *rac*-**2a** under our standard catalytic conditions, when stopped ahead of time (40 min), led to the recovery of unreacted **2a** as a racemic mixture (Scheme 9D, entry 1). This experiment ruled out the intervention of a KR in this reaction. Similar experiments with enantioenriched **2a** did not show any racemization of **2a** during the FC reaction (Scheme 9D, entries 2–3), indicating that dynamic kinetic resolution (DKR) is not operative either.

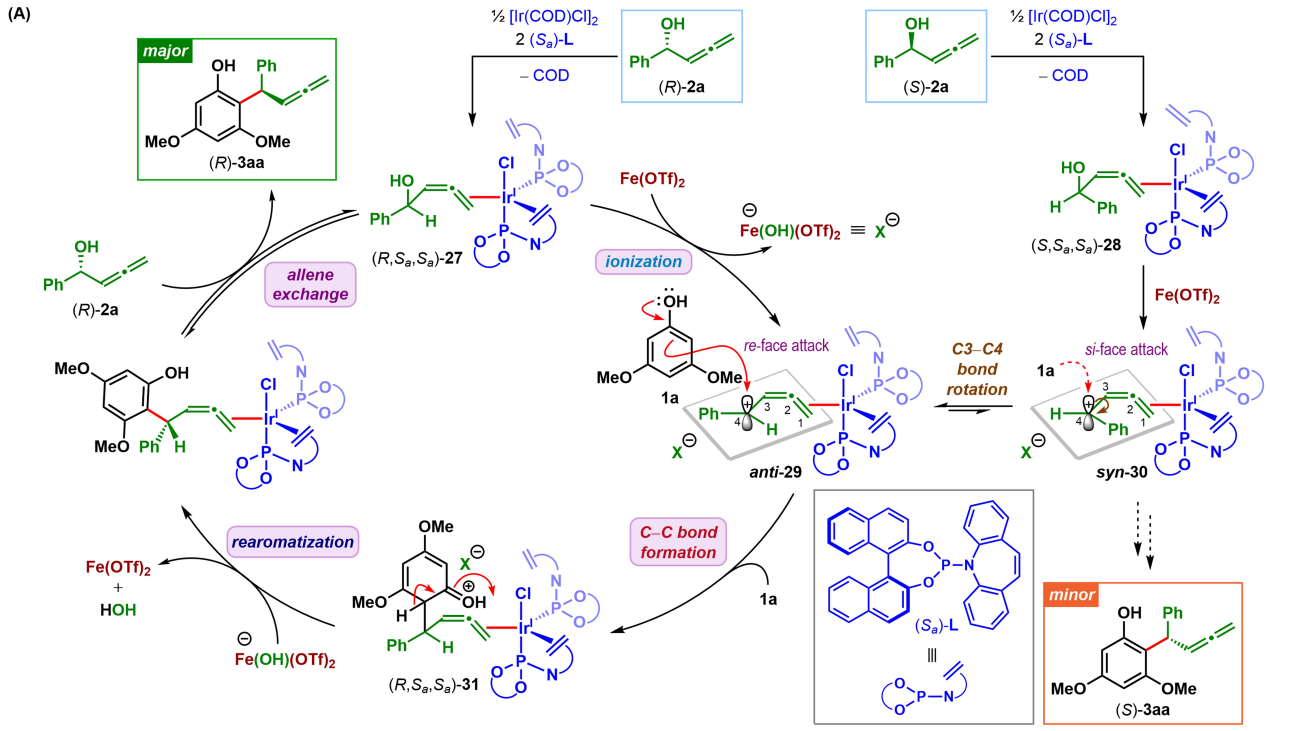
When using (S_a)-**L**, the bottom faces of the planar cationic butadienylium fragment in *anti*-**29** and *syn*-**30** are hindered by the axially bound binaphthyl moiety on Ir(I) (Scheme 9A).[15] Therefore, nucleophilic attack of **1a** from the exposed top face (i.e., *re*- in *anti*-**29** and *si*- in *syn*-**30**) should result in (*R*)- and (*S*)-**3aa**, respectively. Between these two intermediates, *syn*-**30** is more congested compared to *anti*-**29**, due to the vicinity of phenyl ring to the sterically hindered Ir-center. Therefore, the reaction is channeled through *anti*-**29**, to furnish (*R*)-**3aa** as the major product. Starting from enantioenriched **2a**, when the FC reaction was carried out using racemic (*P*,olefin) ligand *rac*-**L**, the product **3aa** was formed as a racemic mixture (Scheme 9D, entry 4). Evidently, the stereochemical information present in allenylic alcohol is lost during the reaction, and the product configuration is exclusively dictated by the ligand on iridium. The loss of stereochemical information of the starting allenylic alcohol **2a** can be explained by the facile equilibration between the Ir-bound carbocationic intermediates **29** and **30** through rotation around the C3–C4 bond. The overall reaction represents a dynamic kinetic asymmetric transformation (DyKAT) of the racemic allenylic alcohols.

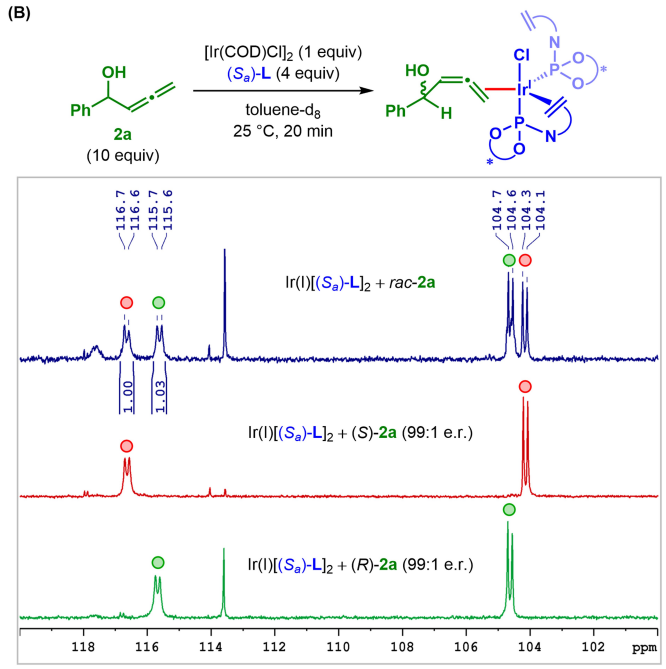
Conclusions

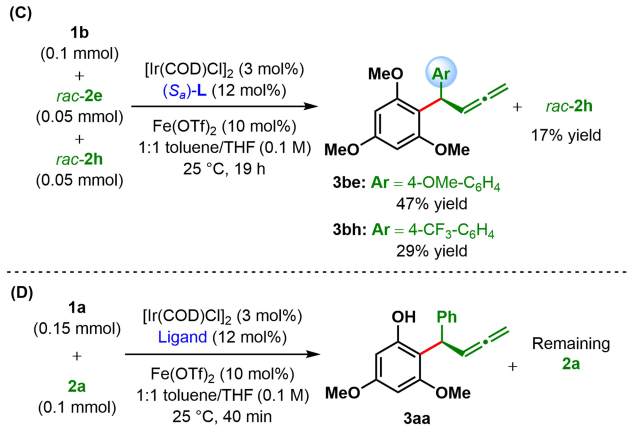
In conclusion, (hetero)arenes were employed for the first time as nucleophiles in Ir-catalyzed allenylic substitution reaction and led to the development of the first catalytic enantioselective Friedel-Crafts allenylation for creating central chirality.[34] Using easily accessible racemic allenylic alcohols as the electrophile, this reaction is catalyzed by a cooperative combination of a Ir(I)-bis(phosphoramidite,olefin) complex and Lewis acidic Fe(OTf)$_2$ and proceeds in an enantioconvergent fashion to deliver 1,1-disubstituted allenylic methanes, generally with good to excellent enantioselectivity. This reaction is completely regioselective in favor of allenes over 1,3-dienes. Hydrogenation of the allene functionality in the products allowed for the overall enantioselective formal introduction of a 4-carbon alkyl chain into (hetero)arenes with the creation of a benzylic stereocenter. While this FC reaction displays complete regiocontrol with respect to (hetero)arenes, it is possible to synthesize a few other regioisomeric FC alkylation products through the judicious choice of (hetero)arene derivatives, post-synthetic modification, or by removal of functional groups from the (hetero)arene ring. In certain cases, it is possible to access

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Entry	Starting material	Ligand	3aa Yield [%]	3aa e.r.	Remaining 2a Yield [%]	Remaining 2a e.r.
1	*rac*-**2a**	(*Sa*)-**L**	47	98:2	37	50:50
2	(*S*)-**2a** (99:1 e.r.)	(*Sa*)-**L**	53	98:2	35	99:1
3	(*S*)-**2a** (99:1 e.r.)	(*Ra*)-**L**	58	2:98	28	99:1
4	(*S*)-**2a** (99:1 e.r.)	*rac*-**L**	56	50:50	26	99:1

Scheme 9. A tentative catalytic cycle for enantioselective Friedel-Crafts allenylation reaction and supporting experiments. In all cases, the yields correspond to the NMR yield as determined by ^{1}H NMR analysis of the unpurified reaction mixture using mesitylene as the standard.

both product enantiomers through the choice of substrate combinations under the influence of the same catalyst. Preliminary experiments established the feasibility of an enantioselective intramolecular FC allenylation using tertiary allenylic alcohols as substrate under the same catalytic conditions. The latter finding is particularly exciting as it leads to the enantioselective construction of a diaryl-containing all-carbon-quaternary center. Experiments towards mechanistic elucidation revealed dynamic kinetic asymmetric transformation (DyKAT) of racemic allenylic alcohols as the reason for enantioconvergency in this FC allenylation reaction.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Friedel-Crafts reaction · iridium catalysis · allenylic substitution · enantioselectivity · (hetero)arenes

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